

WRITTEN EXPLANATION: DIMENSIONAL MODELING STRATEGY FOR PALLADIUM BANK RETAIL ANALYTICS

1. Introduction

The objective of this project was to overhaul Palladium Bank's legacy reporting structure by transforming flat, performance-heavy transaction logs into a scalable, high-performance Star Schema. The primary business driver was the need for the Head of Retail Banking to identify churn signals and customer behavior patterns across 18 months of historical data. By implementing a dimensional model in PostgreSQL, we have transitioned from a simple data storage mindset to a strategic data engineering framework that prioritizes query speed, historical accuracy, and analytical flexibility.

2. DESIGN REASONING: THE STAR SCHEMA ARCHITECTURE

I selected a Star Schema as the foundational architecture, consisting of a central FactTransactions table surrounded by four dedicated dimension tables: DimCustomer, DimBranch, DimProduct, and DimDate.

- **Defining the Grain:** The "grain" of the fact table is set at the individual transaction level. In banking analytics, aggregating data too early (e.g., daily totals) is a common pitfall. By keeping the grain at the transaction level, we allow for "Atomic Data" analysis. This means an analyst can drill down from a high-level national revenue report all the way to a single POS purchase in a specific branch in Lagos.
- **Surrogate Keys vs. Natural Keys:** A critical design choice was the implementation of Surrogate Keys (SK)—integers like Customer_SK. Operational systems often reuse or change IDs (Natural Keys) during system upgrades. By using internal Surrogate Keys, our analytical model is "future proofed." Even if the bank changes its core banking software and assigns a customer a new ID, our data warehouse can map that change without breaking historical links.

3. SLOWLY CHANGING DIMENSIONS (SCD) STRATEGY

One of the most complex requirements for Palladium Bank is tracking changes in customer behavior, specifically "Tier" upgrades (e.g., Silver to Gold). To solve this, I implemented SCD Type 2 for the DimCustomer table.

Unlike SCD Type 1, which simply overwrites old data, SCD Type 2 maintains a full history by using Start_Date, End_Date, and an Is_Current flag. For example, if a customer is upgraded from "Gold" to "Platinum" today, the system marks the Gold record as expired and creates a new Platinum record.

- **Business Justification:** This is essential for accurate attribution. If the bank analyzes revenue from the "Gold" tier for the year 2023, that customer's transactions must be counted toward Gold, not their current Platinum status. This prevents "historical revisionism" and allows for true trend analysis over the 18-month period.

4. ETL/ELT STRATEGY AND DATA QUALITY

The migration of 18 months of history required a structured Extract, Transform, Load (ETL) pipeline. The order of operations was strictly enforced: Dimensions were loaded first to satisfy Referential Integrity constraints, followed by the Fact table.

To maintain the "Gold Standard" of banking data, I implemented four critical Data Quality Guardrails:

1. **Referential Integrity Check:** Ensuring every transaction ID matches an existing customer and branch.
2. **Null Value Scans:** Rejecting any record missing an Amount or Txn_Date.
3. **Range Validation:** Identifying outliers (e.g., a 100-million Naira withdrawal) for manual review.
4. **Deduplication:** Using the Txn_ID as a unique constraint to prevent double-counting transactions during incremental daily loads.

5. HIERARCHIES, PERFORMANCE, AND SCALABILITY

To ensure the model remains performant as the bank grows, I designed specific hierarchies and optimization strategies:

- **Hierarchies:** I defined a Time Hierarchy (Year → Quarter → Month) and a Geographic Hierarchy (State → Branch). These allow stakeholders to "Drill Down" or "Roll Up" data instantly within their BI tools (like Power BI or Tableau).
- **Partitioning:** I recommended Range Partitioning by Month for the FactTransactions table. By splitting the table into monthly chunks, the database can skip millions of irrelevant rows when a user runs a "Last Month" report, drastically reducing CPU load.
- **Indexing:** B-Tree indexes were placed on all Surrogate Keys. This acts as a "Table of Contents" for the database, ensuring that JOIN operations between the Fact and Dimension tables are lightning-fast.

6. CONCLUSION

The resulting Star Schema provides Palladium Bank with a robust, "single source of truth." By separating descriptive attributes (Dimensions) from quantitative measures (Facts), the bank can now answer complex questions—such as "Which product type has the highest churn in Abuja?"—in seconds rather than hours. This architecture is not only built for the current 18-month requirement but is prepared for the next decade of data growth.